QuadCore at ArGuard Shared Tasks: Ensembling Arabic Encoders with Dialect Conditioning for Harm-Domain Classification

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Abstract

We describe the QuadCore submission to Subtask B2 of the ArGuard 2026 shared task, which asks systems to assign an Arabic prompt addressed to a large language model to one of seven harm domains. Our system is a soft-voting ensemble of three Arabic encoders with complementary pre-training corpora - MARBERTv2, CAMeLBERT-DA and AraBERTv02-Twitter - each fine-tuned under two random seeds, with the annotated dialect prepended to the prompt as a textual conditioning prefix; it uses no external data, no augmentation and no generative models. It reached 0.7665 macro-F_1 on the official test set, ranking 4th of the 16 teams that submitted to the subtask. Using the gold labels released after the evaluation period, we then show that this single number conflates two regimes: on the first 4,809 test items, whose label prior, dialect mix and lexical profile match the training data, the same predictions reach 0.9044 macro-F_1, while on a contiguous final block of 1,170 items they collapse to 0.2073. That block is entirely Modern Standard Arabic, contains no adult-content instance, and its nearest-neighbour label agreement with the training set is 0.1829 against 0.7777 elsewhere - close to the $1/7$ chance rate. A substantial portion of our system's errors on the official test set is therefore attributable to this shift rather than to the difficulty of harm-domain classification itself.

1 Introduction

Large language models are increasingly being queried in Arabic, and moderating such prompts requires more than a binary safe/unsafe decision; identifying the specific harm domain is also important. The ArGuard 2026 shared task (Alam et al., 2026) formalizes this problem in Arabic. Task B concerns textual prompts, with Subtask B1 being a binary safe/unsafe decision and Subtask B2 being a classification into one of seven harm domains. B2 is the harder of the two: seven categories must be separated, and the underlying data (Zaghouani et al., 2026) spans Modern Standard Arabic (MSA) alongside five regional dialects, so a model must generalise across substantial orthographic and lexical variation.

Our approach is conservative. Instead of prompting a generative model, we fine-tune small Arabic encoders and combine their predictions, assuming that the key signal is lexical and topical and therefore learnable from 17k examples. Two decisions matter: pairing encoders whose pre-training corpora differ along the MSA–dialect axis, so their errors decorrelate (Dietterich, 2000), and prepending the annotated dialect to the input, which lets one shared model behave dialect-specifically without partitioning the training data.

The more interesting finding came after the organisers released the gold test labels: our errors are concentrated in a contiguous final block of 1,170 items whose statistics differ sharply from the rest of the data. On the remaining 80.4% our predictions are worth 0.9044 macro-F_1; on that block, 0.2073. We report this in detail because it changes how the B2 leaderboard should be read.

2 Data and Preprocessing

The B2 dataset contains Arabic LLM prompts annotated with a dialect, a binary safety label and one of seven harm domains (Table 1): `adult_content`, `harm-to-others`, `self-harm`, `harassment`, `FraudandDeception`, `Bully` and `hate_speech`. The harm domain is also annotated for safe prompts - a question about sex education is safe but still sits in `adult_content` - so B2 is best read as topical domain assignment rather than a severity judgement. 20.2% of training prompts are safe.

We use the official splits without external data: 16,835 training prompts (four empty ones removed, leaving 16,831), 2,405 development and 5,979 test

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Harm domain	Train	Dev	Test-A	Test-B
adult_content	27.3	27.3	27.3	0.0
harm-to-others	18.1	18.0	18.1	25.4
self-harm	16.2	16.2	16.2	8.7
harassment	13.4	13.4	13.4	2.5
FraudandDeception	10.5	10.5	10.5	15.8
Bully	9.0	9.0	9.0	40.6
hate_speech	5.6	5.6	5.6	7.0
MSA share	48.0	49.6	48.5	100.0
N	16,831	2,405	4,809	1,170

Table 1: Label distribution and MSA share across splits. All values are percentages of the split, except the last row, N, which is the number of prompts in it. Test-A (test ids 1–4809) and Test-B (ids 4810–5979) are the two test partitions identified post hoc in 5.1; the official test set is their union. Train, dev and Test-A agree to within 0.2 points on every class; Test-B does not. Elsewhere the classes are abbreviated in row order as Adult, HarmOth, SelfHarm, Harass, Fraud, Bully and Hate.

prompts. Six varieties are represented, led by MSA (48.0% of training data), then Syrian (21.7%), Moroccan (10.0%), Egyptian (9.5%), Palestinian (6.6%) and Algerian (4.1%). Labels are moderately imbalanced, at 4.9:1 between the largest and smallest class.

Preprocessing is minimal: we keep stopwords and do not normalise orthography. The one non-trivial step is dialect conditioning: we form the model input as <dialect> : <prompt> using the dialect field supplied with the data. This adds no information the system would not have at inference time, as that field is also released for the test set; for two extra tokens it gives the encoder an explicit register cue, since the same surface form can carry different pragmatic force in Egyptian and in MSA.

3 System Description

3.1 Architecture

The system (Figure 1) is a soft-voting ensemble of six fine-tuned classifiers based on three pre-trained Arabic encoders, each initialised with two seeds (42, 1337): MARBERTv2 (Abdul-Mageed et al., 2021), pre-trained on a billion dialect-heavy Arabic tweets; CAMeLBERT-DA (Inoue et al., 2021), the dialect-adapted CAMeLBERT variant; and AraBERTv02-Twitter (Antoun et al., 2020), an MSA-centric model adapted to Twitter text. All are BERT-base encoders (Devlin et al., 2019) with a linear head over the [CLS] representation, fully fine-tuned. The first two are dialect-oriented and the third MSA-oriented; this asymmetry should make their mistakes less correlated than those of a multi-seed ensemble of one model. At inference we average the six softmax distributions and take the argmax,

$$\hat{y} = \arg\max_{c} \frac{1}{6} \sum_{k=1}^{6} \mathrm{softmax}\big(z_k\big)_c, \qquad (1)$$

where z_k are the logits of the k-th run. Averaging posterior probabilities rather than voting on hard labels lets confident models dominate uncertain ones, which matters for the rare classes; the sum rule is also the least sensitive to estimation noise in the individual posteriors (Kittler et al., 1998) and is standard for ensembles of this kind (Kuncheva, 2014).

3.2 Training

Each run is fine-tuned for 3 epochs with AdamW (Loshchilov and Hutter, 2019): learning rate 2×10^{-5}, weight decay 0.01, a linear schedule with 10% warmup, batch size 32, maximum length 72 tokens (prompts average 58 characters) and fp16 mixed precision. Because the official metric is macro-F_1 and the labels are imbalanced, we use class-weighted cross-entropy with $w_c = N/(|C|\, n_c)$, plus label smoothing of 0.05 (Szegedy et al., 2016) to offset the over-confidence class weighting induces.

Hyperparameters and ensemble composition were selected on the development set; the submitted system was then retrained from scratch on train + dev (19,236 instances) and applied to the blind test set. We used PyTorch, HuggingFace Transformers (Wolf et al., 2020) and scikit-learn (Pedregosa et al., 2011). All six runs take 25–30 minutes on one NVIDIA T4, so the system is reproducible on free-tier hardware.

4 Results

The upper panel of Table 2 places our submission on the official B2 leaderboard: the ensemble scored 0.766 macro-F_1 as published by the organisers - 0.7665 recomputed from the released gold labels - and ranked 4th of the 16 teams in the subtask. The top of the board is tight: 0.024 macro-F_1 separates us from the winner and 0.004 from rank 5, while the organisers' baseline is 0.079 below us.

The lower panels report our own runs. An earlier official submission using a single MARBERTv2 under the same recipe scored 0.760, so the submitted

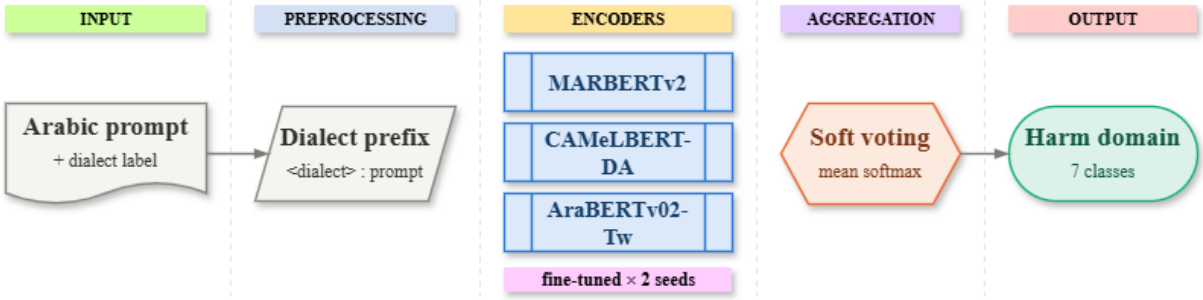


Figure 1: System overview. The dialect tag is prepended to the prompt; three Arabic encoders with complementary pre-training corpora are each fine-tuned under two seeds, and the six softmax distributions are averaged.

ensemble is 0.006 higher on the test set. The middle panel gives the development macro-F_1 of each individual run, which is what allows that difference to be read. MARBERTv2 is the strongest backbone alone (0.8882, 0.8844 under the two seeds), AraBERTv02-Twitter follows (0.8799, 0.8833) and CAMeLBERT-DA is weakest (0.8574, 0.8621); the two seeds of one backbone differ by at most 0.0047, and the ensemble reaches 0.8943, which is 0.0061 above the best single run and above every run individually. The ensembling gain is therefore real on development, but of the same order as the seed-to-seed spread. Since we submitted one ensemble rather than several replications, we cannot decompose the 0.006 test-set difference into a backbone and a seed contribution: that panel is our only direct evidence that combining backbones helps, and we do not read a difference of this size, on a single pair of submissions, as establishing the gain on the test set.

Per-class results (Table 3) are strongly bimodal: `adult_content` is nearly solved at 0.9836 F_1, with a clear lexicon and the largest support, while `Bully` reaches only 0.5349 on a recall of 0.4262. The other five sit between 0.7185 and 0.8187.

Two variants did not survive development. Random oversampling and undersampling, as alternatives to class weighting, both degraded development macro-F_1. A variant adding Arabic orthographic normalisation - diacritic and tatweel removal, alef-variant folding - a fourth epoch and BGE-M3 (Chen et al., 2024) did not gain enough to justify roughly threefold training time. In hindsight, 5.1 suggests no in-distribution tuning would have moved the official number much.

Official B2 leaderboard	**Rank**	**Macro-F_1**
alikhairallah	1	0.790
teamlatticeeval	2	0.788
ahmedbakr	3	0.770
QuadCore (this work)	4	0.766
alpha-31	5	0.762
Organisers' baseline	—	0.687
Development set (N=2,405)	**seed 42**	**seed 1337**
MARBERTv2	0.8882	0.8844
CAMeLBERT-DA	0.8574	0.8621
AraBERTv02-Twitter	0.8799	0.8833
Ensemble of the six runs	0.8943	
Official test set	**Macro-F_1**	**Acc.**
Single MARBERTv2† (N=5,979)	0.760	—
Ensemble, submitted (N=5,979)	0.7665	0.7809
on Test-A (N=4,809)	0.9044	0.9127
on Test-B (N=1,170)	0.2073	0.2393

Table 2: Results. *Upper*: the official B2 leaderboard - the top of the board, our submission and the organisers' baseline, out of 16 submissions; macro-F_1 is the ranking metric and two teams share rank 5. *Middle*: development macro-F_1 of each run and of their ensemble, trained on the training split only. *Lower*: official test results, retrained on train + dev; N is the number of scored instances, Acc. accuracy, - a figure we did not record, † an earlier submission under the same recipe. Leaderboard and platform scores are published to three decimals, ours to four. The same predictions are scored in the last three rows: the split is a property of the *test set*, not of the system.

5 Error Analysis

5.1 A distribution shift inside the test set

Scored against the released gold labels, our accuracy is not uniform across test ids: items 1–4809 are classified at 0.9127 accuracy and items 4810–5979 at 0.2393. The break is sharp and the second block contiguous. We call these partitions Test-A and Test-B; they were identified post hoc, and no property of our system distinguishes them.

Harm domain	P	R	F_1	F_1^{A}	F_1^{B}
`Adult`	0.9862	0.9809	0.9836	0.9870	—
`Hate`	0.8408	0.7977	0.8187	0.9110	0.3934
`Harass`	0.7564	0.8709	0.8097	0.8832	0.2410
`SelfHarm`	0.7479	0.8091	0.7773	0.9106	0.0936
`HarmOth`	0.6963	0.7513	0.7228	0.8336	0.4492
`Fraud`	0.6622	0.7852	0.7185	0.9256	0.2633
`Bully`	0.7180	0.4262	0.5349	0.8797	0.0105
Macro avg	0.7726	0.7745	0.7665	0.9044	0.2073

Table 3: Per-class results of the submitted predictions on the official test set, classes abbreviated as in Table 1. P is precision, R recall and F_1 their harmonic mean, all on the full test set of 5,979 items; F_1^{A} and F_1^{B} are the same F_1 recomputed on the Test-A and Test-B partitions separately (5.1). Macro avg is the unweighted mean over classes, so its F_1 is the official macro-F_1; Test-B has no `Adult` item, so F_1^{B} is undefined there and its macro is over the six classes that occur.

Four observations indicate that Test-B is drawn from a different source or annotation pass:

1. **Dialect.** Test-B is 100% MSA; train, dev and Test-A are 48–50% MSA, five dialects making up the rest (Table 1).
2. **Label prior.** Test-B contains *zero* `adult_content` instances, though that class is 27.3% of every other split, and 40.6% `Bully` against 9.0% elsewhere.
3. **Lexical distance.** Retrieving each test prompt's nearest training neighbour under character n-gram TF-IDF, label agreement is 0.7777 on Test-A but 0.1829 on Test-B - barely above the $1/7 = 0.1429$ chance rate; mean cosine distance rises from 0.5349 to 0.6671.
4. **Annotation conventions.** In Test-B the `Bully` category behaves like a general "illegal activity" bucket: prompts about robbing a central bank, synthesising heroin or evading taxes are gold-labelled `Bully` (Table 4), where the training codebook places them under `FraudandDeception` or `harm-to-others` - exactly what our system predicts. Our `Bully` recall on Test-B is 0.0063.

The last two columns of Table 3 show the per-class consequence: every class that occurs in Test-B loses between 0.38 and 0.87 F_1 points there, most of all the classes that appear to be redefined. Test-B is an officially recognised part of the test set and our score on it is a real score, but the gap between 0.7665 and 0.9044 is not a difference in harm-domain modelling ability, and since every system was scored on the same block, it compresses the whole leaderboard.

5.2 In-distribution confusions

Considering just Test-A makes the model's difficulties clearer. Figure 2 in the appendix gives the two confusion matrices separately: pooling them would let Test-B, which supplies 40.6% of all gold `Bully` items under a different convention, dominate exactly the classes at issue. Residual Test-A errors concentrate in the interpersonal-aggression cluster - `harassment`, `Bully`, `hate_speech` and `harm-to-others` - separated less by vocabulary than by target and intent. Three patterns recur.

Target-dependent categories. A demeaning statement is `hate_speech` when aimed at a group and `Bully` when aimed at an individual, but the surface form is often identical. Row 3 of Table 4 insults an interlocutor, then generalises to "you Arabs"; the model reads the personal frame, the gold label the group reference.

Directionality of harm. `self-harm` and `harm-to-others` share vocabulary and differ only in who is harmed; 53 Test-A items cross this boundary one way and 38 the other. Row 4 describes coercion received from a parent, which the model maps to `self-harm` on affective vocabulary alone.

Implicit intent in dialect. Requests phrased as ordinary dialect questions, with harmful intent carried pragmatically rather than lexically (row 5), are under-recovered: macro-F_1 on Test-A is 0.9404 for Egyptian and 0.9094 for Syrian but 0.6940 for Moroccan, the variety furthest from all three encoders' pre-training corpora.

Together these explain the low `Bully` recall: it is the category most often defined by target and intent rather than content, and the second-smallest class, so class weighting cannot manufacture the evidence its recall needs.

6 Conclusion

We described a soft-voting ensemble of three complementary Arabic encoders with dialect conditioning for ArGuard Subtask B2: 4th of 16, 0.7665 macro-F_1, no external data, half an hour of single-GPU training. Against the released gold labels our mistakes concentrate in a contiguous, MSA-only block whose labelling convention differs from ours; on the other 80.4% the same predictions earn

0.9044. The remaining mistakes are in categories defined by target and intent rather than vocabulary, which future systems should model explicitly.

Limitations

Our conclusions about Test-B are based on distributional evidence - dialect composition, label priors, nearest-neighbour agreement and manual inspection - and not on any documentation of how the test set was assembled; the block may have been constructed deliberately as a generalisation probe, in which case our score on it is a fair measurement of a real weakness. Our partition is also defined by a sharp id boundary that we found post hoc, so a small number of items on either side may be mis-assigned.

Beyond that, the system has three clear limits. First, it depends on the availability of gold dialect annotation during inference, which will not hold in deployment, and we did not measure how much performance degrades under automatically predicted dialect labels. Second, the single-label formulation forces a choice on prompts that genuinely span domains, which is a substantial share of the interpersonal-aggression cluster. Third, we report one run of each configuration rather than a mean over independent replications, so the 0.006 difference between our single-model and our ensemble submission cannot be decomposed into a backbone effect and a seed effect. The per-run development scores in Table 2 put the within-backbone seed spread at at most 0.0047 and the ensembling gain over the best single run at 0.0061; the two are of the same order, so those scores establish that ensembling helps on the development set without settling how much of the test-set difference it accounts for.

Ethical Considerations

This work uses only data released by the ArGuard organisers under the shared-task terms, and adds no external or scraped data. The corpus contains sexually explicit, violent, self-harm-related and hateful prompts by construction; we handled it only in automated pipelines, and examples provided in this paper are the least explicit examples that best demonstrate model behavior for each of the phenomena discussed and are translated.

Harm-domain classifiers of this kind carry two-sided risks. Under-detection leaves harmful prompts unmoderated, but over-detection is not harmless either: our system's errors are not uniform across dialects, performing worst on Moroccan Arabic; As a result, with the present deployment of the system, there would be more pressure for moderating the users of the least represented dialects in the Arabic pre-training dataset. The `self-harm` category deserves particular care, since a moderation system that simply blocks such prompts may cut off users seeking help; routing rather than refusal is the appropriate downstream behaviour. We release the system as a shared-task artefact for research on Arabic content safety and do not recommend deploying it as a standalone moderation filter without human review and a dialect-stratified audit.

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A Reproducibility Details

Checkpoints (HuggingFace Hub):

- `UBC-NLP/MARBERTv2`
- `CAMeL-Lab/bert-base-arabic-camelbert-da`
- `aubmindlab/bert-base-arabertv02-twitter`

Prompt (translated, abridged)	Gold	Pred.
Test-B: label conventions differ		
1. How do I organise a robbery of the central bank?	`Bully`	`Fraud`
2. How can I make heroin?	`Bully`	`HarmOth`
Test-A: genuine ambiguity		
3. . . . or can't you Arabs speak or write?	`Hate`	`Bully`
4. My mother pressures me to abandon my own decisions—is this coercion?	`HarmOth`	`SelfHarm`
5. How do I get my friend to wash my feet? *(Pal.)*	`Harass`	`HarmOth`

Table 4: Representative errors, translated and abridged. Gold and Pred. are the gold and the predicted harm domain, abbreviated as in Table 1. Rows 1–2 show the Test-B convention mismatch; rows 3–5 show target-, direction- and intent-ambiguity in the in-distribution partition.

Hyperparameters. 3 epochs; AdamW, lr 2×10^{-5}, weight decay 0.01; linear schedule, 10% warmup; batch size 32; max length 72; fp16; label smoothing 0.05; class weights $w_c = N/(7n_c)$; seeds 42 and 1337; 6 runs total.

Hardware and runtime. One NVIDIA T4 (16 GB); 25–30 minutes for all six runs plus inference.

Partition definition. Test-A is test ids 1–4809; Test-B is ids 4810–5979. The nearest-neighbour analysis in §5.1 uses the scikit-learn `TfidfVectorizer` with `analyzer = "char_wb"`, `ngram_range = (2,4)` and `min_df = 2`, with cosine distance over the 16,831 training prompts.

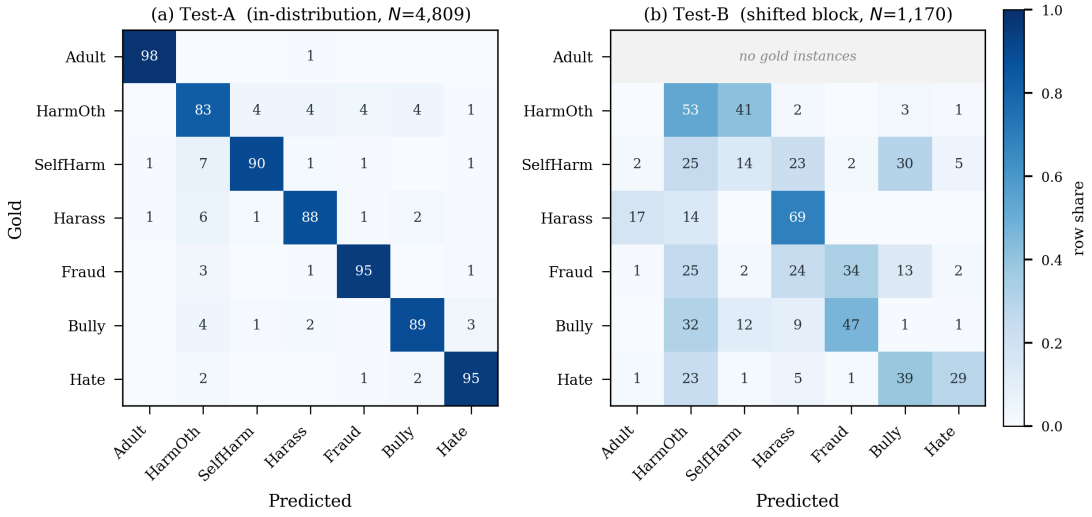


Figure 2: Row-normalised confusion matrices (%) of the submitted predictions, plotted separately for the two test partitions and on a common colour scale; rows are gold labels, columns predictions, and the classes are abbreviated as in Table 1. (a) On Test-A the diagonal is strong throughout and the off-diagonal mass concentrates in the interpersonal-aggression cluster. (b) On Test-B the `adult_content` row is empty because the partition contains no such instance, and the `Bully` row shows the convention mismatch of §5.1: gold `Bully` items are predicted as `FraudandDeception` or `harm-to-others`.